Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [20ME33]

THIRDSEMESTER B.Tech EXAMINATION, DECEMBER–2021

FLUID MECHANICS

Time:3 Hours **Maximum Marks:70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	State Newton's law of viscosity. Discuss the effect of viscosity with temperature in case of liquids and gases.	**07**
	b.	The dynamic viscosity of oil used for lubrication between a shaft and sleeve is 6 poise. The shaft is of diameter 0.4 m and rotates at 190 r.p.m. Calculate the power lost in the bearing for a sleeve length of 90mm. The thickness of the oil film is 1.5mm.	**07**
2.	**a.**	Prove that centre of pressure for a vertical surface submerged in fluid lies below the centre of gravity of that surface and is independent of the density of the liquid	**07**
	b.	Calculate the total pressure on a circular plate of 1.5m diameter which is placed vertically in water in such a way that the centre of the plate is 3m below the surface of the water. Also, find the position of centre of pressure.	**07**
3.	**a.**	What is Euler's equation of motion? How will you obtain Bernoulli's equation from it?	**07**
	b.	A pipe of diameter 400mm carries water at a velocity of 25m/s. The pressures at the points A and B are given as 29.43N/cm^2 and 22.563N/cm^2 respectively while the datum head at A and B are 28m and 30m . Find the loss of head between A and B	**07**

OR

4.	**a.**	An oil of Sp.gr 0.8 is flowing through a venturimeter having inlet diameter 20 cm and throat diameter of 10cm. The oil-mercury differential manometer shows a reading of 25 cm. Calculate the discharge of oil through the horizontal venturimeter. Take Cd=0.98.	**07**
	b.	What is a Pitot tube? How will you determine the velocity at a point with the help of a pitot tube?	**07**
5.	**a.**	Derive an expression for discharge through fully submerged orifice.	**06**
	b.	Derive an expression to find the rate of flow through convergent-divergent mouth piece.	**08**

OR

6. a. Derive the expression for discharge over a triangularnotch. **08**

b. A rectangular notch 2m wide has a constant head of 500mm. Find the discharge over the notch, if the coefficient of discharge over the notch is 0.62. Also if the rectangular notch has a discharge of 0.24 m^3/s with head of 800mm, find its length. Assume Cd for the second scenario to be 0.6 **06**

7. a. Derive an expression for the loss of head due to friction in the pipes (Darcy-Weishbach equation) **08**

b. In a pipe of diameter 350mm and length 75m water is flowing at a velocity of 2.8m/s Find the head loss due to friction using Darcy-Weishbach equation and Chezy's formula for which C=55. Assume kinematic viscosity of water as 0.012 stoke **06**

OR

8. a. Explain the following terms; **06**

i. Hydraulic gradient line

ii. Total energy line

iii. Water hammer in pipe

b. A horizontal pipe of 150mm dia is joined by sudden enlargement to a 225mm dia pipe. Water is flowing at a rate of 0.05 m^3/s. Find the loss of head due to abrupt expansion, pressure difference in two pipes and change in pressure if change of section is gradual without any loss. **08**